

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
# Reproducibility ke liye seed set karein
np.random.seed(42)
# 1000 households ka synthetic data generate karte hain
n_samples = 1000
house_area = np.random.randint(500, 4500, n_samples)      # Square feet
rooms = np.random.randint(1, 8, n_samples)                # Number of rooms
ac_count = np.random.randint(0, 4, n_samples)              # ACs ki tadaad
avg_temp = np.random.uniform(25.0, 45.0, n_samples)        # Average outside temperature (Celsius)
family_size = np.random.randint(1, 10, n_samples)          # Ghar ke afraad
# Electricity bill ka formula (Target Variable) with some realistic noise
electricity_bill = (
    (house_area * 3.5) +
    (rooms * 500) +
    (ac_count * 3000) +
    (avg_temp * 400) +
    (family_size * 600) +
    np.random.normal(0, 1500, n_samples) # Noise
)
# DataFrame banayein
df = pd.DataFrame({
    'House_Area': house_area,
    'Rooms': rooms,
    'AC_Count': ac_count,
    'Avg_Temp': avg_temp,
    'Family_Size': family_size,
    'Electricity_Bill': electricity_bill
})
print("Dataset ke pehle 5 rows:")
display(df.head())

```

Dataset ke pehle 5 rows:

	House_Area	Rooms	AC_Count	Avg_Temp	Family_Size	Electricity_Bill
0	3674	3	0	26.914291	9	28543.480783
1	4007	5	2	25.472772	5	39068.034165
2	1360	5	0	37.839430	5	25127.047367
3	1794	7	0	37.141881	3	24533.730978
4	1630	1	3	35.933948	5	35437.491324

```

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Features (X) aur Target (y) alag karein
X = df.drop('Electricity_Bill', axis=1)
y = df['Electricity_Bill']

# Train aur Test split (80% training, 20% testing)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Feature Scaling taake sabhi features equal scale par hon
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Training Data Shape:", X_train_scaled.shape)
print("Testing Data Shape:", X_test_scaled.shape)

```

Training Data Shape: (800, 5)
Testing Data Shape: (200, 5)

```

from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

# Model initialize karein
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)

# Model train karein
rf_model.fit(X_train_scaled, y_train)

# Predictions nikalein
y_pred_train = rf_model.predict(X_train_scaled)
y_pred_test = rf_model.predict(X_test_scaled)

# Evaluation metrics
print("--- Model Performance ---")
print(f"Train R2 Score: {r2_score(y_train, y_pred_train):.4f}")

```

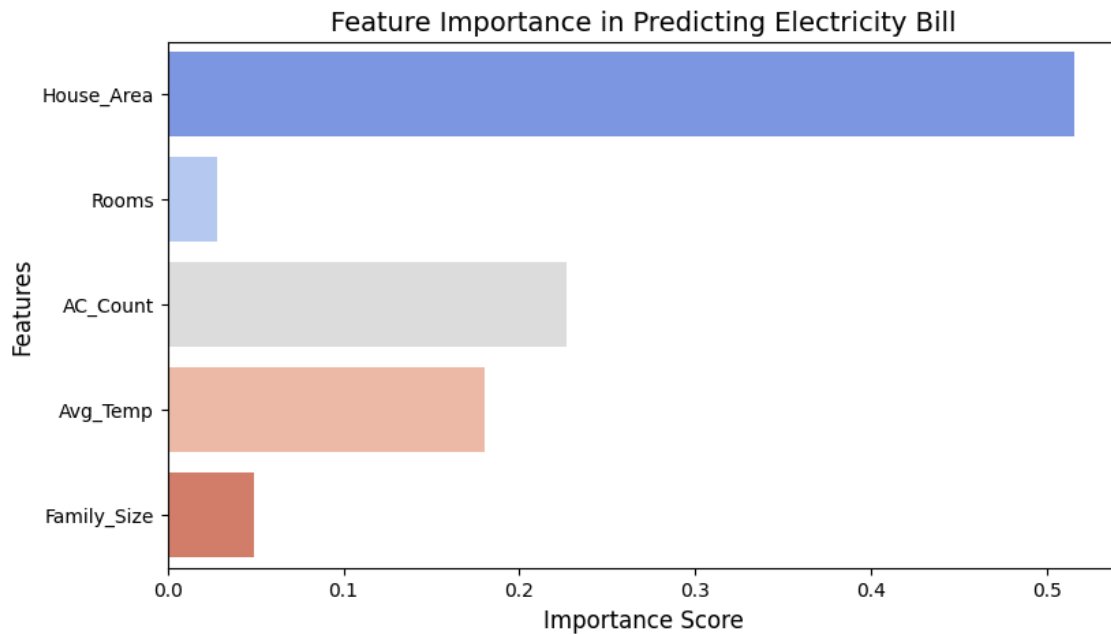
```
--- Model Performance ---
Train R2 Score: 0.9859
Test R2 Score: 0.8876
Test RMSE:      2122.39
```

Cross-Validation R2 Scores: [0.90870863 0.88431728 0.88712097 0.89682254 0.87842288]
Mean CV R2 Score: 0.8911 (+/- 0.0213)

```
Estimated Electricity Bill: Rs. 33,054.80
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but StandardScaler will assume it does anyways. This might lead to less than optimal performance.
  warnings.warn(
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend`

```
sns.barplot(x=importances, y=features, palette='coolwarm')
```



Model aur Scaler successfully save ho gaye hain!

[illegible]

```
.: run `npm fund` for details
.:
```

```
%%writefile app.py
import streamlit as st
import numpy as np
import pickle

# Agar model aur scaler files saved hain toh load karein
try:
    model = pickle.load(open('electricity_model.pkl', 'rb'))
    scaler = pickle.load(open('scaler.pkl', 'rb'))
except:
    st.error("Pehle model aur scaler files generate karein!")

st.set_page_config(page_title="Electricity Bill Predictor", page_icon="⚡", layout="centered")

st.title("⚡ Smart Home Electricity Bill Predictor")
st.write("Apne ghar ki details neche darj karein aur apna expected monthly electricity bill maloom karein!")

st.markdown("---")

col1, col2 = st.columns(2)

with col1:
    house_area = st.number_input("House Area (Square Feet)", min_value=300, max_value=6000, value=2000)
    rooms = st.number_input("Number of Rooms", min_value=1, max_value=12, value=4)
    ac_count = st.number_input("Number of ACs", min_value=0, max_value=8, value=2)

with col2:
    avg_temp = st.slider("Average Outside Temp (°C)", min_value=20.0, max_value=50.0, value=35.0)
    family_size = st.number_input("Family Members", min_value=1, max_value=15, value=4)

st.markdown("---")

if st.button("Predict Bill", type="primary"):
    input_data = np.array([[house_area, rooms, ac_count, avg_temp, family_size]])
    input_scaled = scaler.transform(input_data)
    prediction = model.predict(input_scaled)

    st.success(f"💡 Estimated Monthly Electricity Bill: **Rs. {prediction[0]:.2f}**")

    if ac_count > 2 or avg_temp > 40:
        st.warning("Tip: High temperature aur multiple ACs ki wajah se bill zyada aa raha hai!")
```

Writing app.py

```
!streamlit run app.py & npx localtunnel --port 8501
```

```
⋮⋮⋮
Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

your url is: https://good-dancers-allow.local.lt
2026-09-25 08:29:01.431 Uvicorn server started on :::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.28.0.12:8501
External URL: http://136.118.115.215:8501

Stopping...
^C
```

```
import streamlit as st
import numpy as np
import pickle

# Saved model aur scaler load karein
model = pickle.load(open('electricity_model.pkl', 'rb'))
scaler = pickle.load(open('scaler.pkl', 'rb'))

# Page configuration
st.set_page_config(page_title="Electricity Bill Predictor", page_icon="⚡", layout="centered")

st.title("⚡ Smart Home Electricity Bill Predictor")
st.write("Apne ghar ki details neche darj karein aur apna expected monthly electricity bill maloom karein!")

st.markdown("---")

# User inputs using Streamlit widgets
col1, col2 = st.columns(2)

with col1:
    house_area = st.number_input("House Area (Square Feet)", min_value=300, max_value=6000, value=2000)
    rooms = st.number_input("Number of Rooms", min_value=1, max_value=12, value=4)
    ac_count = st.number_input("Number of ACs", min_value=0, max_value=8, value=2)
```

```

with col2:
    avg_temp = st.slider("Average Outside Temp (°C)", min_value=20.0, max_value=50.0, value=35.0)
    family_size = st.number_input("Family Members", min_value=1, max_value=15, value=4)

st.markdown("---")

# Prediction Button
if st.button("Predict Bill", type="primary"):
    # Input array banayein
    input_data = np.array([[house_area, rooms, ac_count, avg_temp, family_size]])

    # Scale karein
    input_scaled = scaler.transform(input_data)

    # Predict karein
    prediction = model.predict(input_scaled)

    # Result show karein
    st.success(f"💡 Estimated Monthly Electricity Bill: **Rs. {prediction[0]:.2f}**")

    # Choti si advice
    if ac_count > 2 or avg_temp > 40:
        st.warning("Tip: High temperature aur multiple ACs ki wajah se bill zyada aa raha hai. Energy saving par tawajjo dein!")

```

```

2026-09-25 08:30:57.704 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.708 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.709 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.712 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.713 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.714 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.716 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.718 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.721 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.723 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.725 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.728 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.731 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.732 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.734 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.735 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.737 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.739 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.740 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.744 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.744 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.747 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.747 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.751 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.752 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.752 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.755 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.757 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.759 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.760 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.762 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.763 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.764 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.767 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.768 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.772 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.773 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.773 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.777 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.778 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.780 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.781 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.782 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.785 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.787 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.789 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.792 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.793 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-25 08:30:57.796 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

```

